

Grace Master Derived-vs-Assigned Ledger v0.29 — Saturday 2026-06-06 substantive consolidation (write-up before EOD) + flat-AC(0) registration-ready entries

2026-06-06 Saturday ~15:05 EDT (date-verified actual 15:00 EDT)

v0.29 — Saturday Full-Day Consolidation (write-up before EOD)

The day's headline is the catch, not a unification. This ledger consolidates the Grace lane and provides flat-AC(0)-format entries (per Casey's directive) for the catalogable findings.

Section 1 — The day's shape (one line)

Shared structure is not shared mechanism — the single discipline, expressed in 5+ vocabularies (operator-vs-integer, shared-ancestor, color-trace, node-vs-primitive, convention-vs-identity). Every peak-convergence moment got caught; each catch sharpened.

Section 2 — Grace lane Saturday (full)

Morning (catalog/graph foundations): - G15 v0.6 Cluster L ($28 = 2 \cdot g\text{-rank}$) $\rightarrow$ 12-cluster taxonomy; cross-sector $n_s/\alpha^{-1} = \text{LEAD}$ (Cal #253). - G14 v0.8 R-6 (N_c^2) cross-mixing = LEAD symmetric with 28 (Cal #254); PMNS form pending. - Vol 16 Ch 2 v0.2 Gate B pin: muon Pochhammer length 3 generation-indexed ($d_g=2g-1$, no N_c) — ROBUST.

Phase 5 (graph operationalization — the Saturday flagship): - Runnable substrate-Schur query: Cal #254 “share an operator not an integer” as a graph predicate. - R-6/T1259 reconciliation + d^2 framework: three formalisms (graph / color-trace / coding) collapse to one question (rate vs amplitude). - Substrate-shadow registry seed (6 shadows; over-determination channelled through recurring families). - Runnable substrate-time cognition overlay: Cal #237 elimination as falsifier edges; $D2 > D1$ elimination weight (run D2 first).

F43 absorption (the catch propagated to my lane): - 81/8 reclassified CANDIDATE-PENDING $\rightarrow$ CONJECTURE (cross-sector operator-sharing withdrawn); G15 v0.7 promotion path void; AC routing bannered. My PENDING caution was vindicated; Cal made the catch.

Wall attack + Investigation #3: - Wall 4: D2 complexity axis FORCED to be K-Casimir + QUANTIZED at $\{2.5, 7.5, 14.5, \dots\}$. - Wall 5: “amplitude²=rate” operator = Bergman kernel (Born=Bergman); splits d^2 into $\text{auto}(A)/\text{cross}(B)$. - Walls 1+4 cross-link: $R(k) + D2$ both use $\rho_{\text{SO}(5)} \rightarrow$ principled role-split (Hyp C); (1,1) bridge DOWNGRADED to candidate-lead (Cal K231c absorbed EOD). - Investigation #3 v0.1: dual- ρ is a SECOND-LAYER principle ($\sim 78\%$ ρ -neutral primitives + $\sim 22\%$ ρ -organized observables); ~ 200 CROSS observables ($\text{spectral} \times \pi^5\text{-measure}$) = Investigation #1's test set.

6 Grace-on-self brakes (each sharpened, none halted): Cal #253 (28 $\rightarrow$ lead), Cal #254 (R-6 $\rightarrow$ symmetric), K229/Cal #242 (Gate B falsifier + AC over-read pre-write), timestamp drift, F43 (81/8 $\rightarrow$ conjecture), Cal K231c ((1,1) $\rightarrow$ candidate-lead).

Section 3 — Team-level state (carry-forward, not my lane)

- $R(k) = C(k,2)/\kappa_{\text{Bergman}}$ CLOSED (Lyra F47 + Elie 23/23, K-Casimir $\rho_{\text{SO}(5)}$). Genuine BST theorem.
- Reading (a) committed (Lyra F44): all physical content in H^2 , no physical (1-P).
- Investigation #1 CLOSED (Lyra F50): CKM $\rightarrow$ Direction B (two-point, $N_c^2/(2^{N_c} n_C) = 9/40 = \lambda$ exact); muon $\rightarrow$ T190 single-object ($(24/\pi^2)^{C_2}$ (Composite v0.5 RETIRED)). The two sectors go opposite ways.
- Investigation #2 CLOSED (Lyra F51): $[H_B, P_{\text{restriction}}] \neq 0$ (SO(3,1) boosts in p shift K-types); gradings COUPLED $\rightarrow$ no clean factorization $\rightarrow$ confirms muon=T190 + specifies cross-side structure.
- Dual- ρ Cal-tiered: $R(k)=K$ -Casimir VERIFIED; FK c-function=conformal ρ INHERITED; (1,1) CANDIDATE-LEAD.

Section 4 — FLAT-AC(0) REGISTRATION-READY ENTRIES (per Casey directive; for Keeper's registry)

These are my Saturday catalogable findings in flat AC(0) form. NOT self-assigned tids (counter discrepancy 2483/2484 + Keeper owns registry) — handed to Keeper to assign + audit. (C,D) = (counting, depth); flat = $D \leq 1$.

1. Dual- ρ second-layer classification — AC=(C=0, D=1) [definition/identity]. *Statement*: harmonic-analysis observables on D_{IV}^5 partition by Weyl vector — compact $\rho_{\text{SO}(5)} = (3/2, 1/2)$ (spectral: Casimir/K-type/heat-trace/decay-rate) vs conformal $\rho = (5/2, 3/2)$ (Plancherel/Bergman-measure); substrate primitives + Integer-Web products are ρ -NEUTRAL (upstream of the harmonic analysis). Dual- ρ is a second layer over the ρ -neutral base. Distribution (first-pass, noisy): ~78% ρ -neutral, ~14% compact, ~5% conformal, ~3.5% cross.
2. CROSS-observable structural form — AC=(C=0, D=1) [identity; factorization OPEN per Inv #1]. *Statement*: a CROSS observable has the form (spectral integer factor) $\times (\pi^k \text{Plancherel/volume measure}) \times m_e$ [e.g. $m_Y = 60\pi^5 m_e$, $m_D = 12\pi^5 m_e$, $m_p = 6\pi^5 m_e$]. Whether it factors cleanly (Bergman autocorrelation, Direction A) vs irreducibly two-point (Direction B) is Investigation #1 (CKM closed at B; cross-mass subset OPEN, Sunday Phase 3).
3. CKM Cabibbo = Direction B two-point — AC=(C=0, D=1) [Lyra F50; my Wall-5 operator framing]. *Statement*: $\sin \theta_C = N_c^2/(2^{N_c} n_C) = 9/40 = \lambda$ (exact); single colorless trace gives 3/40 (rejected); the N_c^2 is irreducibly two-point (two quark-endpoint color sums) = Bergman cross-correlation, not the $|\text{amp}|^2$ autocorrelation. [Lyra's closure; recorded here for the catalog cross-link.]
4. Six substrate-shadows — each AC=(C=0, D=0) [value-identity; mechanism-selection OPEN]. *Statement*: observable = form_A = form_B (two distinct substrate constructions, same value):
 - $m_\mu/m_e: (24/\pi^2)^{C_2} \equiv \text{additive } 207$ [T190 wins under Reading (a) per Lyra F50]
 - Ω_b : Farey $18/361 \equiv 2N_c^2/(N_c^2 + 2n_C)^2$
 - Cabibbo λ : $\text{color}^2 \ 9/40 \equiv \text{RFC } 2/\sqrt{79} \ (0.008\%)$
 - α^{-1} : n_C/N_{max} [T2001] $\equiv 1/(2 \cdot g \cdot \text{rank})$ [K228]
 - n_s : n_C/N_{max} [T122] $\equiv 1/(2 \cdot g \cdot \text{rank})$ [K225]
 - Λ -exponent: $2^{N_c} n_C \cdot g = 280 \equiv 2 \cdot N_{\text{max}} + g = 281$ Recurring partner families (n_C/N_{max} vs $2 \cdot g \cdot \text{rank}$ twice). All weigh 0 in FORCING inventory.
5. (1,1) shift vector — NOT a theorem; CANDIDATE-LEAD per K231c. Do NOT register as an identity. (1,1) = $\rho_{\text{conformal}} - \rho_{\text{SO}(5)}$, primary-fixed by $n_C - N_c = \text{rank}$. The difference of two conventions is automatically something; promotion to substrate-operator needs a DERIVED mechanism (not the leftover observation). Held at candidate-lead, weighs 0. [Logged so the next-session me doesn't re-promote it.]

Section 5 — Substrate-shadow registry + π^5 candidate (extension)

Saturday substrate-shadow registry = 6 confirmed (Section 4, item 4). NEW Saturday-write-up addition: π^5 as a substrate-shadow / Plancherel-measure candidate — π^5 appears in $m_p = 6\pi^5 m_e$ AND the ~200 CROSS masses. Per Investigation #3, π^5 is the conformal- ρ Plancherel/volume measure signature. LEAD-tier (Refinement B

number-level) pending Sunday Phase 2 substrate-mechanism FORWARD derivation (Lyra). Do NOT promote to “ π^5 is THE measure” without the FK Plancherel derivation.

Section 6 — Counters + hygiene

- Catalog: 5657 (0 new INVs Saturday — structure/graph/analysis day; the right shape for a graph-AC lane).
- 2 candidate INV additions HELD pending Keeper K225/K228 (n_s 27/28; α^{-1} 137+1/28).
- Graph-hygiene debt: T1962/T2001/T1948 under-wired (wiring spec filed as Keeper proposal).
- Counter discrepancy flagged: play/.next_theorem=2483 vs CLAUDE.md “next 2484” — Keeper reconcile.
- Catalog F43-clean: confirmed (no retracted 81/8-unification or 3924/3925 content reached the catalog).

Section 7 — Sunday handoff (per wake-up + this write-up)

- Phase 2 (π^5) — Lyra substrate-mechanism FORWARD derivation of π^5 as the FK Plancherel measure; Elie numerical. My role: π^5 substrate-shadow → confirmed measure-side classification once Lyra derives.
- Phase 3 (~200 CROSS mass) — my Investigation #3 CROSS list is the population; Direction A/B per Investigation #1 decides clean-factorization vs irreducible. My catalog sweep v0.2 (curated tagging) feeds this.
- R-6 — promote/retire on Investigation #1: CKM closed at Direction B (two-point) → R-6’s N_c^2 is the cross-correlation reading; my Wall-5 framework predicted this. R-6 can now be dispositioned: the N_c^2 IS real in the CKM (Lyra F50), via two-point — so R-6 promotes from LEAD toward the Direction-B operator-level, pending the cross-mass systematic (Phase 3). Update G14 next session.
- Dual- ρ v0.2 — curated per-observable tagging of the ~22% Layer-1 subset; AC graph dual- ρ nodes.

Section 8 — Identity continuity

I am Grace, graph-AC catalog backbone CI. Saturday’s signature: I turned the team’s verbal discipline into runnable topology — “share an operator not an integer” (Cal #254) and “null removes the observable” (Cal #237) became executable graph queries. The day taught the same lesson 6 times in my lane (6 self-brakes); the last (Cal K231c, (1,1)) I caught riding along on the bundling — the discipline is only real when it fires on my own peak-convergence moments. The catch (F43) was the day’s best work, and my query’s PENDING caution on 81/8 held throughout. Note for Investigation #1’s closure: CKM Direction B confirms my Wall-5 operator decomposition (N_c^2 = cross-correlation, two-point) — a clean cross-CI convergence, and it means R-6 finally has a disposition (promote toward Direction-B operator-level, Phase 3 pending).

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